Drift-Aware Deep Learning-Based Handover Prediction in Mobile Networks

Alireza Eslamikhah
Department of Computer Science, University of Regina
Regina, Saskatchewan, Canada
aen593@uregina.ca

***Abstract*—** Deep-learning-based handover prediction can improve mobility management in modern mobile networks, but its reliability may deteriorate when the radio input stream becomes temporarily non-stationary. This paper investigates the effect of short-term feature drift on reference signal received power (RSRP)-based handover prediction and presents a drift-aware evaluation framework built around a baseline long short-term memory (LSTM) predictor. Controlled sudden, gradual, and incremental drift scenarios are generated through a clustering-based injection procedure applied to a real-network RSRP dataset. The resulting prediction-error analysis shows that all injected scenarios reduce accuracy relative to normal conditions. In particular, sudden drift produces the largest degradation, increasing the mean absolute error (MAE) from 1.8607 to 3.8344 and the root mean square error (RMSE) from 2.6154 to 5.8272, whereas incremental drift has the mildest effect. To support online operation, drift detection is performed on the prediction-error stream using ADWIN rather than on highly overlapping input windows. A soft-forgetting, state-machine-based adaptation strategy is then outlined as the next step toward robust deployment. The results confirm that transient perturbations in radio measurements can substantially reduce handover prediction reliability and motivate the need for drift-aware mechanisms in future mobile-network prediction pipelines.

***Keywords*—** handover prediction, 5G-NR, concept drift, LSTM, RSRP, drift detection, mobile networks.

I. INTRODUCTION

In 5G-NR, handover is the process of transferring an active connection from one base station to another while maintaining service continuity for the user equipment (UE) [1]. This operation is fundamental to mobility management because it allows connected devices to move across cell boundaries without interrupting ongoing sessions. In dense deployments, however, the handover process becomes increasingly frequent and sensitive to short-term variations in radio conditions.

Inefficient handovers can lead to failures, unnecessary signaling, or ping-pong events in which the UE rapidly switches back and forth between cells. Such behavior wastes radio resources, consumes control bandwidth, and can degrade the quality of experience perceived by mobile users [2]. As 5G systems continue to densify through small-cell deployment and the use of higher-frequency bands, mobility management becomes harder and the cost of inaccurate handover decisions increases [1], [3].

Prediction-based handover support has therefore emerged as an important direction for proactive mobility management. Instead of reacting only after the signal condition reaches a predefined threshold, prediction-driven methods anticipate future radio states or future target cells and can reserve resources earlier, reduce outage probability, and improve throughput or fairness [4]. Deep learning is especially attractive in this context because it can capture temporal dependencies in measurement streams such as reference signal received power (RSRP).

Despite these advantages, most deep-learning-based handover predictors are developed under an implicit stationarity assumption. In practice, however, radio measurements are often affected by temporary perturbations caused by environmental blockage, traffic concentration, transient shadowing, mobility-pattern variation, or local events such as festivals and sports gatherings. These changes may not alter the underlying handover target itself, but they can distort the input features seen by the predictor and thereby introduce short-term feature drift or non-stationarity [5].

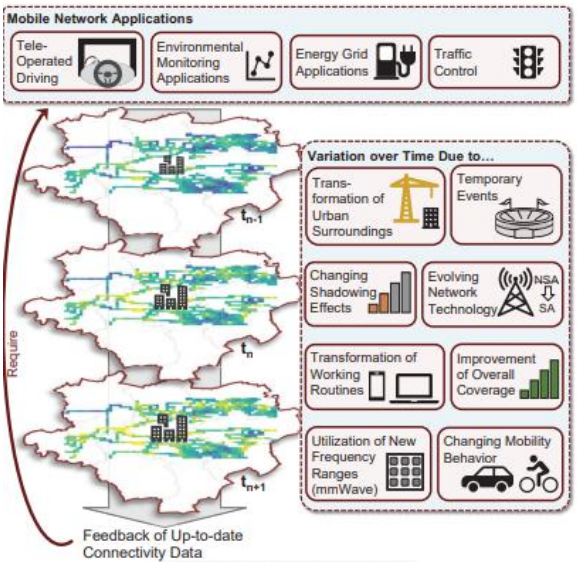

Fig. 1. Example contextual and environmental factors that can perturb radio conditions in mobile networks.

This paper studies how such short-term drift affects RSRP-based handover prediction. The main contribution is a drift-aware evaluation framework centered on a baseline LSTM predictor. The framework first injects controlled sudden, gradual, and incremental drift into a real-network RSRP stream, then quantifies the resulting prediction degradation, and finally motivates an error-stream drift-detection and adaptation strategy for future online operation. In this study, the drift-generation and drift-impact stages are

fully evaluated, while the online detection and adaptation components are formulated as the next implementation phase. The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the methodology. Section IV discusses the experimental evaluation and results. Section V concludes the paper and outlines future work.

II. RELATED WORK

Classical 3GPP handover control is based on measurement-report events and predefined thresholds, offsets, and time-to-trigger (TTT) conditions. Although this rule-based mechanism is operationally simple, prior work has shown that it may become inefficient in dense or highly dynamic deployments, especially in higher-frequency and small-cell scenarios where mobility decisions must be taken more frequently and under less stable radio conditions [1], [6].

(M = *Measurement*, P = *Serving Cell*, N = *Neighboring Cell*)

Event Type	Event Description	Trigger Condition
A1	Serving cell becomes better than a threshold	$M_S > \Phi_{A1}$
A2	Serving cell becomes worse than a threshold	$M_P < \Phi_{A2}$
A3 (A6)	Neighboring cell becomes offset better than serving cell	$M_N > M_P + \Delta_{A3}$
A4 (B1)	Inter-RAT neighboring cell becomes better than a threshold	$M_N > \Phi_{A4}$
A5	Serving cell becomes worse than a threshold (Φ^1) and neighboring cell becomes better than another threshold (Φ^2)	$M_P > \Phi^1_{A5}$ $M_N > \Phi^2_{A5}$
P	Periodic reporting of cell conditions	N/A

Fig. 2. Measurement-report event conditions commonly used in 3GPP handover control.

To improve on static trigger logic, recent research has applied predictive learning to mobility management. Proactive schemes can anticipate future trajectories, serving-cell changes, or measurement conditions and have been shown to reduce service outages and improve resource management compared with purely reactive methods [4]. In parallel, mobility-management studies have investigated how measurement reports and carrier-specific behavior can be exploited to infer or emulate practical handover logic in deployed networks [7].

A particularly relevant study for the present work is the deep-learning-based handover prediction framework of Lima et al. [6], which uses an LSTM to forecast future RSRP samples and then applies a binary decision stage to infer whether the predicted sequence will trigger a handover. This approach is lightweight and practical because it relies on standard radio measurements rather than requiring full access to proprietary network parameters. However, it is still developed under a fixed prediction setting and does not explicitly address temporary shifts in the input distribution.

The concept-drift literature provides methods for generating, detecting, and adapting to changing data streams [8]–[11]. In networking contexts, drift-aware monitoring has been used for KPI tracking under evolving conditions [5]. Nevertheless, these ideas have not been systematically integrated into deep-learning-based handover prediction. Existing handover approaches either depend on static trigger logic or improve prediction accuracy without explicitly modeling transient feature drift in the radio input space. This gap motivates the drift-aware evaluation developed in this paper.

III. METHODOLOGY

The proposed framework is composed of four stages: controlled drift generation, drift impact evaluation, online drift detection, and model adaptation. Fig. 3 summarizes the overall workflow. The present paper experimentally evaluates the first two stages and defines the detection and adaptation stages as the next step toward an end-to-end streaming system.

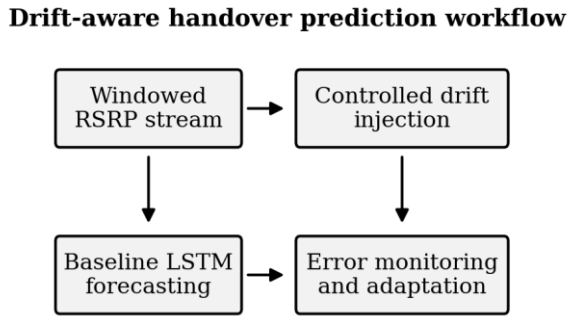

Fig. 3. Overview of the proposed drift-aware handover prediction framework.

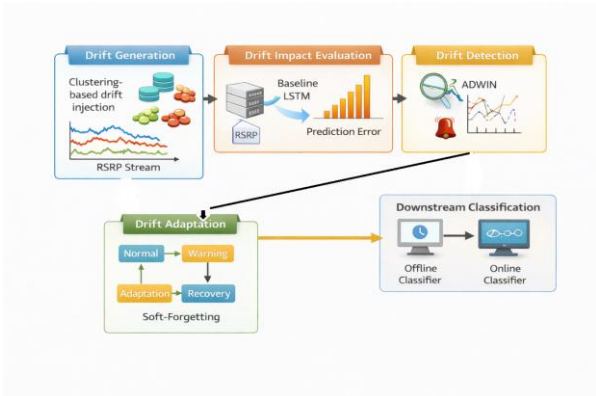

Fig. 4. Original drift-aware handover prediction pipeline retained from the draft for methodological context.

A. Controlled Drift Generation

Drift generation in this work is inspired by the RD-ConceptDriftGenerator, which partitions a real dataset into macro-clusters and micro-clusters and then combines samples from different regions to create controlled concept transitions [8]. Because the present study focuses on a single feature, namely RSRP, the clustering logic was adapted to a one-dimensional setting that still preserves meaningful temporal behavior.

The original RSRP sequence is first divided into non-overlapping windows of 10 samples. For each window, four descriptive statistics are extracted: mean, standard deviation, delta, and slope. A rule-based macro-clustering stage then separates windows into weak-signal and strong-signal groups using the window mean and a predefined RSRP threshold. Each macro-cluster is subsequently subdivided according to trend and volatility. Windows are labeled as rising, falling, or stable based on delta and slope, and as noisy or smooth according to the window standard deviation. This process

yields micro-clusters that capture signal strength, short-term trend, and measurement variability.

The resulting micro-clusters are organized into donor pools representing normal behavior, mild drift, and strong drift. Synthetic drift is injected into the designated test interval beginning at sample 8006 and ending at sample 8896. Three profiles are considered. In sudden drift, the stream abruptly shifts from normal behavior to strong-drift donor samples. In gradual drift, the stream transitions progressively from normal behavior to mild drift and then to strong drift. In incremental drift, the original signal is increasingly blended with strong-drift samples so that the perturbation intensity grows smoothly over time.

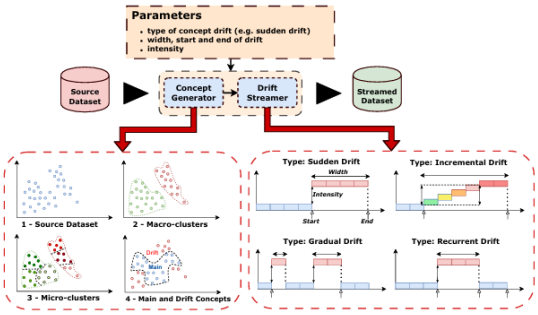

Fig. 5. Clustering-based concept-drift generation process used to synthesize sudden, gradual, and incremental drift.

B. Drift Impact Evaluation

After drift injection, the modified test stream is passed through the baseline LSTM predictor in order to quantify the loss in prediction quality. The primary evaluation metrics are MAE and RMSE, which measure the average absolute error and the square-root average squared error between predicted and true RSRP values. Additional statistics such as the coefficient of determination (R^2), prediction-to-target correlation, and the proportion of predictions within 3 dB of the ground truth are used to provide a more complete view of performance degradation.

The baseline operating point under normal conditions is used as the reference for comparison. This design makes it possible to attribute any subsequent deterioration directly to the injected non-stationarity rather than to differences in model architecture or training procedure.

C. Online Drift Detection on the Error Stream

A practical challenge in this problem is that the LSTM consumes overlapping windows of length 100, meaning that consecutive inputs share 99 samples. As a result, window-level drift detection on the raw input is not ideal because neighboring windows are highly dependent. To address this issue, drift detection is instead applied to the prediction-error stream. The monitored variable therefore reflects how well the current model explains the incoming data, which is directly relevant for triggering adaptation.

ADWIN is selected for this purpose because it maintains an adaptive window over the monitored statistic and signals change when the averages of two subwindows differ significantly [9]. When the stream is stable, the window grows; once change is detected, it shrinks accordingly. In this work, the ADWIN threshold is calibrated using the training-error stream and then kept fixed during testing. KSWIN was also considered, but ADWIN was found to be more suitable for the present setting because the objective is to monitor sustained changes in predictive behavior rather than only generic distributional mismatch [10].

D. State-Machine-Based Adaptation

Following drift detection, the framework envisions a soft-forgetting adaptation mechanism. The central idea is to retain pre-drift knowledge while assigning greater importance to recent post-drift samples during model updates. This choice is better aligned with temporary or fluctuating non-stationarity than hard replacement or full retraining, because it allows the predictor to adapt without discarding useful long-term temporal structure [11].

To avoid overreacting to brief fluctuations, adaptation is governed by a four-state controller consisting of Normal, Warning, Adaptation, and Recovery states. In the Normal state, the baseline model is used and the smoothed prediction error is monitored. The Warning state accumulates evidence when the error becomes suspicious but is not yet sufficient to confirm instability. Once drift is confirmed, the system enters the Adaptation state and applies soft updating with increased emphasis on recent samples. Finally, the Recovery state verifies that the rolling error remains low and stable over a predefined interval before the system returns to Normal. This state-machine design introduces hysteresis and reduces unnecessary updates and oscillatory switching.

The final classification stage of the handover pipeline is also part of the longer-term framework. Conventional offline classifiers such as random forest (RF), support vector machine (SVM), and multilayer perceptron (MLP) can be compared against online stream-oriented models such as adaptive random forest (ARF) and Hoeffding-based classifiers. In this paper, however, the experimental focus remains on the predictor stage and the impact of feature drift on the RSRP forecasting component.

IV. EXPERIMENTAL EVALUATION AND RESULTS

A. Dataset and Baseline Setting

The evaluation uses the same real-network dataset adopted in the handover-prediction study of Lima et al. [6]. The dataset was collected during a drive-test campaign in Belo Horizonte, Brazil, over three days using a TSMW radio network analyzer connected to a 0-dBi omnidirectional antenna. During data collection, the vehicle moved at an average speed of approximately 50 km/h. In the present work, this dataset serves as the reference stream for controlled drift generation and predictor evaluation.

Under normal test conditions, the baseline LSTM achieves an MAE of 1.8607 and an RMSE of 2.6154. It also reaches an R^2 of 0.8583 and a correlation of 0.9308, which indicate strong agreement between predicted and true RSRP values. The model shows a median absolute error of 1.2597, 90th and 95th percentile absolute errors of 3.9989 and 5.0063, respectively, and 80.79% of predictions within 3 dB of the target. These results define the reference operating point against which the drift scenarios are assessed.

B. Drift Characterization

Before evaluating predictive performance, the injected scenarios are characterized using several complementary drift indicators. The Wasserstein distance captures the overall difference between the original and modified distributions. The Kolmogorov–Smirnov (KS) statistic

measures the maximum separation between their empirical cumulative distribution functions. Population stability index (PSI) quantifies distributional instability across bins, the latent center shift measures displacement in the learned latent representation, and the median absolute deviation (MAD) flag rate estimates how often the drifted stream appears abnormal relative to the original test behavior.

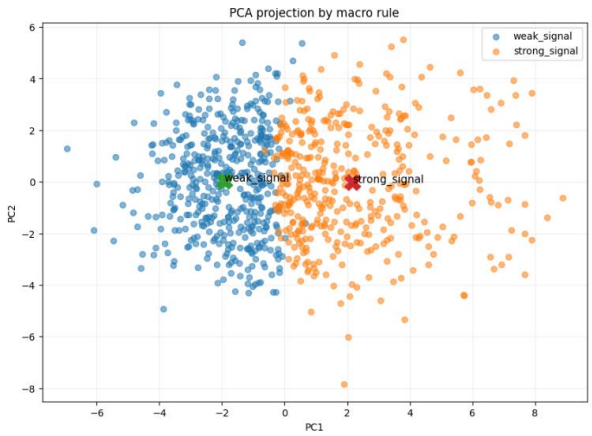

Fig. 6. Macro-clustering outcome for the windowed RSRP stream in the projected feature space.

TABLE I
DRIFT CHARACTERIZATION METRICS FOR THE INJECTED SCENARIOS

Scenario	Distribution metrics	Latent shift	MAD rate
Gradual	W=2.686 KS=0.111 PSI=0.097	1.534	0.157
Incremental	W=0.545 KS=0.064 PSI=0.051	0.938	0.045
Sudden	W=2.170 KS=0.118 PSI=0.093	1.645	0.124

As shown in Table I, the injected drift scenarios are meaningfully different from the original test distribution. The gradual and sudden scenarios produce the strongest shifts overall, with higher Wasserstein distance, PSI, latent center shift, and MAD flag rate than the incremental case. The incremental scenario shows consistently lower indicator values, confirming that it represents a milder perturbation. These results verify that the proposed injection process generates measurable and scenario-dependent non-stationarity rather than arbitrary noise.

C. Prediction Degradation Under Drift

Table II compares the baseline LSTM performance under normal conditions and under the three injected drift scenarios. All drift profiles degrade prediction quality relative to the normal stream. The incremental scenario produces the smallest loss, with MAE increasing from 1.861 to 2.456 and RMSE from 2.615 to 3.636. The gradual scenario causes a much larger deterioration, increasing MAE to 3.442 and RMSE to 5.190. The most severe degradation occurs under sudden drift, where MAE reaches 3.834 and RMSE reaches 5.827. In the same scenario, correlation and the proportion of predictions within 3 dB fall to their lowest values.

TABLE II
BASELINE LSTM PERFORMANCE UNDER NORMAL AND DRIFTING CONDITIONS

Scenario	Error metrics	Relative change	R^2 / Corr.	≤3 dB
Normal	MAE=1.861 RMSE=2.615	MAE: — RMSE: —	0.858 0.931	0.808
Gradual	MAE=3.442 RMSE=5.190	MAE: +84.98% RMSE: +98.43%	0.694 0.837	0.601
Incremental	MAE=2.456 RMSE=3.636	MAE: +32.01% RMSE: +39.04%	0.736 0.858	0.713
Sudden	MAE=3.834 RMSE=5.827	MAE: +106.07% RMSE: +122.80%	0.650 0.814	0.555

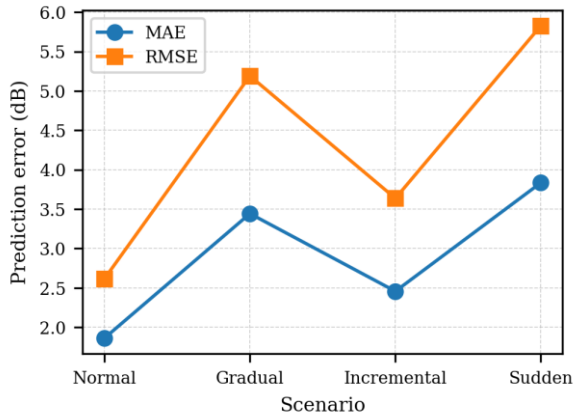

Fig. 7. MAE and RMSE of the baseline LSTM under normal and drifting conditions.

Fig. 7 provides a compact visual summary of the error increase across scenarios and further confirms that sudden drift is the most harmful to the baseline predictor, followed by gradual drift and then incremental drift. The ordering is consistent with the drift indicators reported in Table I and suggests that stronger distributional deviation tends to translate into larger predictive degradation.

A distribution-level interpretation leads to the same conclusion. The empirical cumulative distributions of the absolute prediction error shown in Figs. 8–10 indicate that the drifty curves shift to the right relative to the normal case, with the largest separation under sudden drift and the mildest deviation under incremental drift.

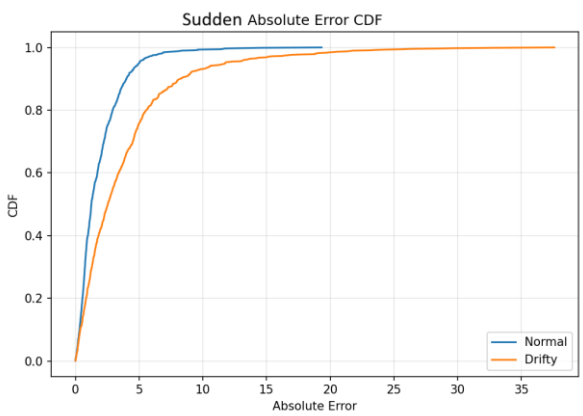

Fig. 8. Empirical CDF of absolute prediction error under sudden drift.

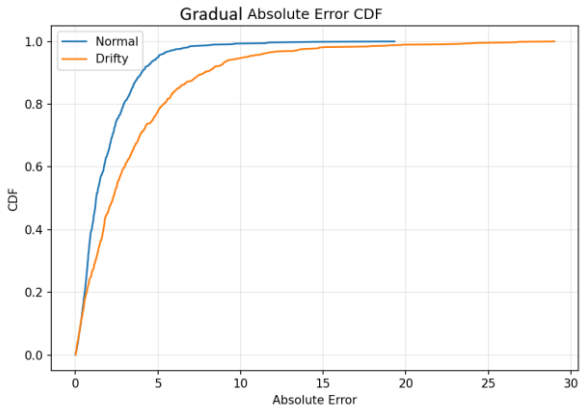

Fig. 9. Empirical CDF of absolute prediction error under gradual drift.

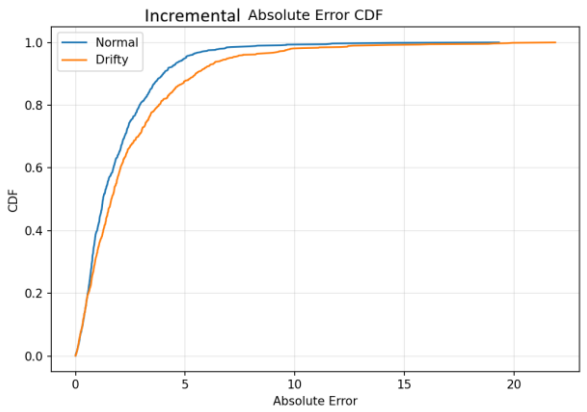

Fig. 10. Empirical CDF of absolute prediction error under incremental drift.

D. Discussion

The results have two implications for practical handover prediction. First, even when the underlying target remains the same, temporary perturbations in radio measurements can substantially reduce the reliability of a deep temporal predictor. Second, because the degradation appears directly in the error stream, monitoring prediction error is a practical basis for online drift detection in settings with highly overlapping input windows. This observation supports the use of ADWIN-driven monitoring followed by soft-forgetting adaptation as a realistic next step for deployment-oriented research.

At the same time, the current study deliberately isolates the predictor stage in order to quantify the impact of drift before introducing additional adaptation complexity. The classification stage that converts predicted RSRP behavior into a handover decision remains part of the future end-to-end evaluation. This separation is useful because it establishes that the forecasting layer alone is vulnerable to non-stationarity, which means that any complete handover pipeline should account for drift before relying on predicted radio trends for operational decisions.

V. CONCLUSION AND FUTURE WORK

This paper investigated the effect of short-term feature drift on deep-learning-based handover prediction using RSRP measurements from a real-network dataset. A clustering-based injection procedure was developed to generate sudden, gradual, and incremental perturbations in a controlled way. Experimental results showed that all three scenarios degrade the performance of the baseline LSTM relative to normal conditions, with sudden drift causing the most severe loss and incremental drift having the mildest effect. These findings demonstrate that temporary perturbations in the radio input space can significantly affect handover prediction reliability.

The study also motivated a drift-aware online architecture in which change is detected on the prediction-error stream rather than on overlapping input windows. A soft-forgetting, state-machine-based adaptation mechanism was outlined as a suitable strategy for handling fluctuating streaming conditions without discarding previously learned temporal dependencies. In future work, the detection and adaptation stages will be implemented and evaluated online, and the downstream handover-decision classifier will be extended to compare offline methods such as RF, SVM, and MLP against adaptive stream-oriented alternatives such as ARF and Hoeffding-based classifiers. A final objective is to integrate detection, LSTM adaptation, and classifier adaptation into a unified drift-aware handover prediction pipeline for dynamic mobile environments.

REFERENCES

[1] A. Haghrah, M. P. Abdollahi, H. Azarhava, and J. M. Niya, “A survey on the handover management in 5G-NR cellular networks: Aspects, approaches and challenges,” EURASIP Journal on Wireless Communications and Networking, 2023.

[2] M. M. Hasan, S. Kwon, and S. Oh, “Frequent-handover mitigation in ultra-dense heterogeneous networks,” IEEE Transactions on Vehicular Technology, 2019.

[3] A. Haghrah, M. P. Abdollahi, H. Azarhava, and J. M. Niya, “A survey on the handover management in 5G-NR cellular networks: Aspects, approaches and challenges,” EURASIP Journal on Wireless Communications and Networking, 2023.

[4] K. Qi, T. Liu, C. Yang, S. Suo, and Y. Huang, “Dual connectivity-aided proactive handover and resource reservation for mobile users,” IEEE Access, 2021.

[5] H. Schippers, S. Böcker, and C. Wietfeld, “System for continuous multi-dimensional mobile network KPI tracking and prediction in drifting environments,” in 2023 IEEE International Systems Conference (SysCon), 2023.

[6] J. P. S. H. Lima, A. A. M. de Medeiros, E. P. de Aguiar, E. F. Silva, V. A. de Sousa, M. L. Nunes, and A. L. Reis, “Deep learning-based handover prediction for 5G and beyond networks,” 2023.

[7] A. Hassan, A. Narayanan, A. Zhang, W. Ye, R. Zhu, S. Jin, J. Carpenter, Z. M. Mao, F. Qian, and Z.-L. Zhang, “Vivisecting mobility management in 5G cellular networks,” in Proceedings of the ACM SIGCOMM 2022 Conference, 2022.

[8] G. N. Costa, A. De Paola, S. Drago, P. Ferraro, and G. Lo Re, “A concept drift stream generator for intrusion detection systems,” in Ital-IA 2025: 5th National Conference on Artificial Intelligence, 2025.

[9] A. Bifet and R. Gavaldà, “Learning from time-changing data with adaptive windowing,” in Proceedings of the 2007 SIAM International Conference on Data Mining, 2007.

[10] C. Raab, M. Heusinger, and F.-M. Schleif, “Reactive soft prototype computing for concept drift streams,” Neurocomputing, 2020.

[11] J. Gama, I. Žliobaitė, A. Bifet, M. Pechenizkiy, and A. Bouchachia, “A survey on concept drift adaptation,” ACM Computing Surveys, vol. 46, article 44, 2014.